Dabbling with Desmos: When Polar and Cartesian Graphs Collide

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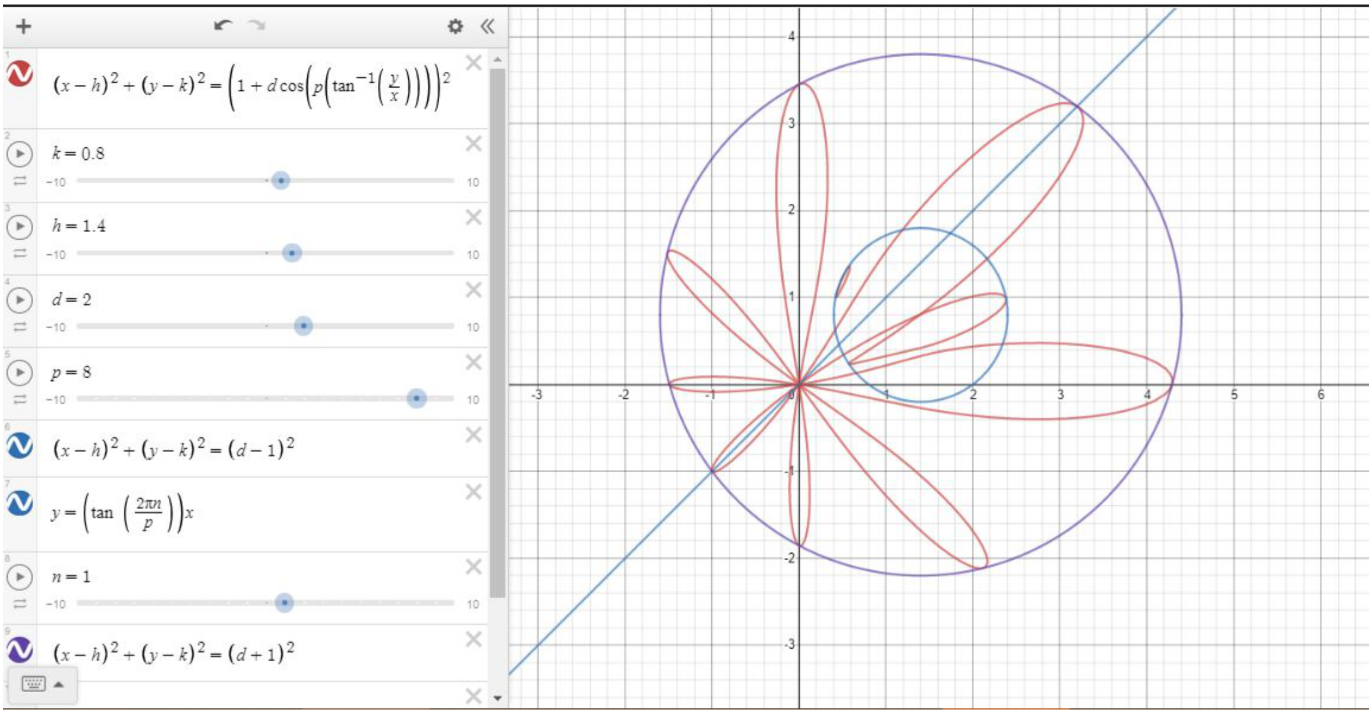
Some of the most beautiful graphs in math are undoubtedly polar equations. I discovered this inadvertently, as I was trying to create a flower-like image on Desmos, the online graphing calculator. Looking for a method to wrap the periodicity of a sine graph radially around a circle, I first tried replacing both the x and the y in the equation of a circle with sine expressions. Though understandably this did not work, that was the point at which I realized I needed to use polar coordinates.

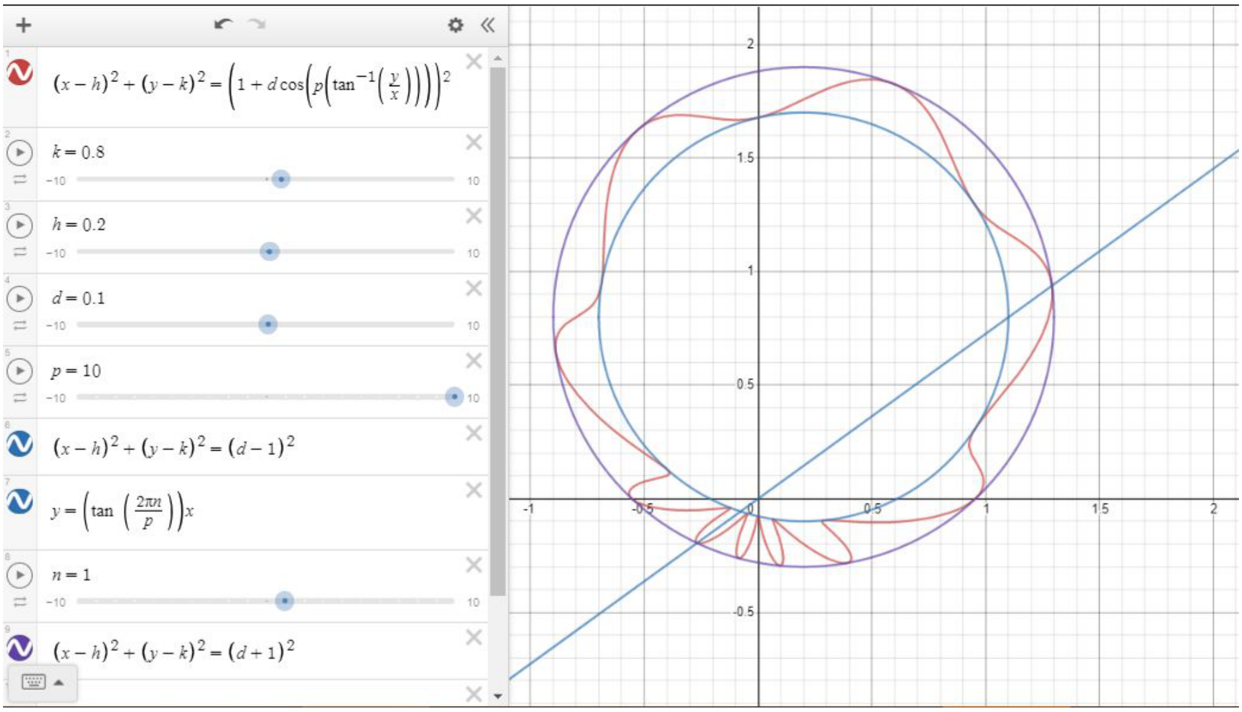
Polar coordinates, in contrast to Cartesian coordinates, are determined by a point's distance from the origin, and angle relative to the positive x-axis. As a result of their radial nature, they lend themselves to rotation. However, since the Cartesian system lends itself better to translation, orthogonal reflection, and dilation, there is benefit in combining the two. Thus, my first thought after creating a flower in polar coordinates was to convert it to Cartesian coordinates. This process led me to the creation of two intriguing graphs, as I experimented with a variety of mathematical transformations.

1 The Shifting Flower

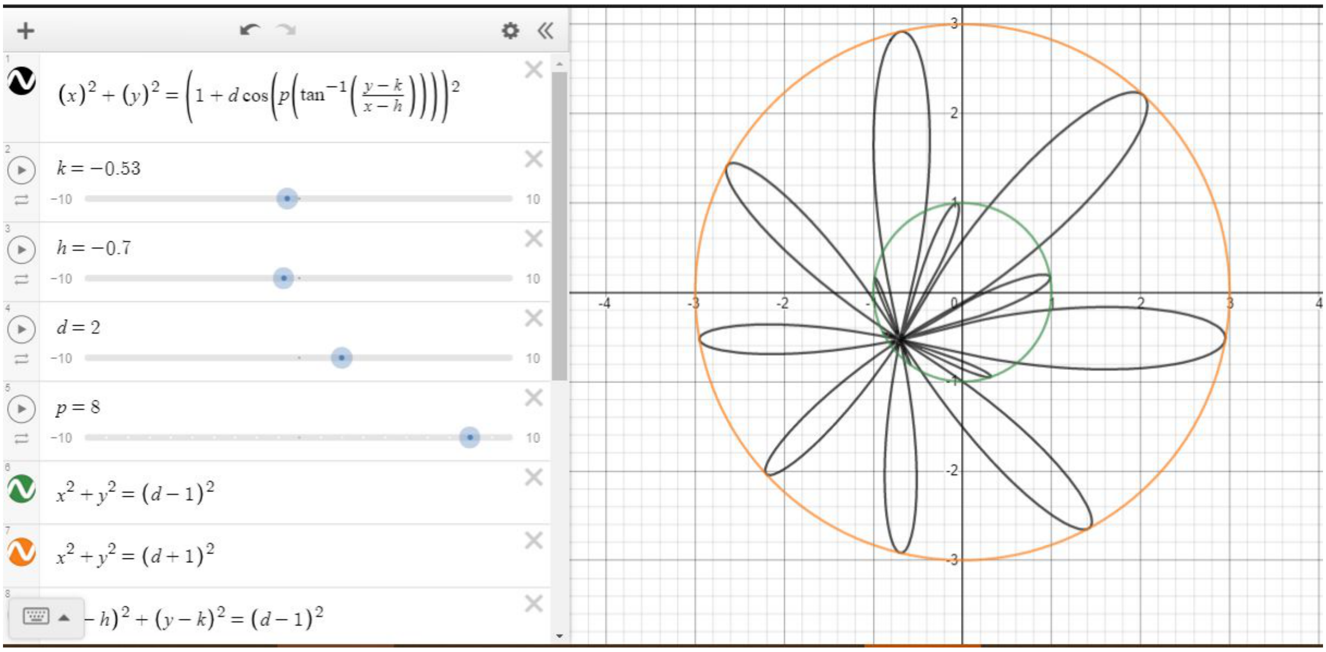
In order to convert polar coordinates to Cartesian coordinates, I replaced r^2 with x^2+y^2, and I wrote θ as $\tan^{-1}\frac{y}{x}$, to get the equation $x^2+y^2=(1-d(\cos(p\tan^{-1}\frac{y}{x}))^2$. From here, I started replacing x^2+y^2 with $(x-h)^2+(y-k)^2$, in an attempt to translate the flower. However, before I could replace $\tan^{-1}\frac{y}{x}$ with $\tan^{-1}\frac{y-k}{x-h}$, I realized that the graph I had created looked nothing like a flower.



The first thing I noticed about this graph was its uneven petals; provided that the graph was sufficiently close to the origin, the shape had petals. Interestingly, regardless of the value of k, h or d, the petals seemed to point in the same direction, as long as p and a were constant, and they always extended from the origin. This can be thought of in terms of the two main elements of the graph. Looking at the left side of the equation, the coordinate plane can be divided radially into p sections. In each section, $d(\cos(p\tan^{-1}\frac{y}{x})$ should run through the same set of values, ranging from $-d$ to d depending on the angle. Without translations, this would be expressed as an adjustment to a point's distance from the origin, or to the radius of the circle, forming a midline of sorts around the graph. However, after this circle is translated, it becomes a distinct second element; adjustments are being made to a point's distance from (h, k), but these adjustments have nothing to do with the point's angle relative to (h, k). As a result, the entire graph can be contained within two circles, $(x - h)^2 + (y - k)^2 = (1 - d)^2$ and $(x - h)^2 + (y - k)^2 = (1 - d)^2$.



This process begs the question of whether graphs can be partially translated. In this graph, only one feature of the graph was moved, while the other remained the same. A very similar graph to this one can be created by replacing $\tan^{-1} \frac{y}{x}$ with $\tan^{-1} \frac{y-k}{x-h}$ while keeping $x^2 + y^2$ the same — in this case, while the point from which the petals extend may move, the circle they are enclosed in remains at the origin.



Another feature of this graph is its two sets of petals. As d nears 1 then exceeds it, $(x-h)^2 + (y-k)^2 = (1-d)^2$ becomes a point, and then begins to expand again. This is because when $d > 1$, it is possible for $(1 - d(\cos(p\tan^{-1}\frac{y}{x}))$ to become a negative value, with a minimum of $1-d$. Graphically, when a point is negated, it is inverted — wherever a point might have been relative to (h,k) along a given line, it is now in the opposite direction from along the same line. As a result, the points in a graph with $d < 1$ that were between petals and tangent to $(x-h)^2 + (y-k)^2 = (1-d)^2$ now become the tips of another set of petals extending from (h,k) with a maximum radius of $d-1$.

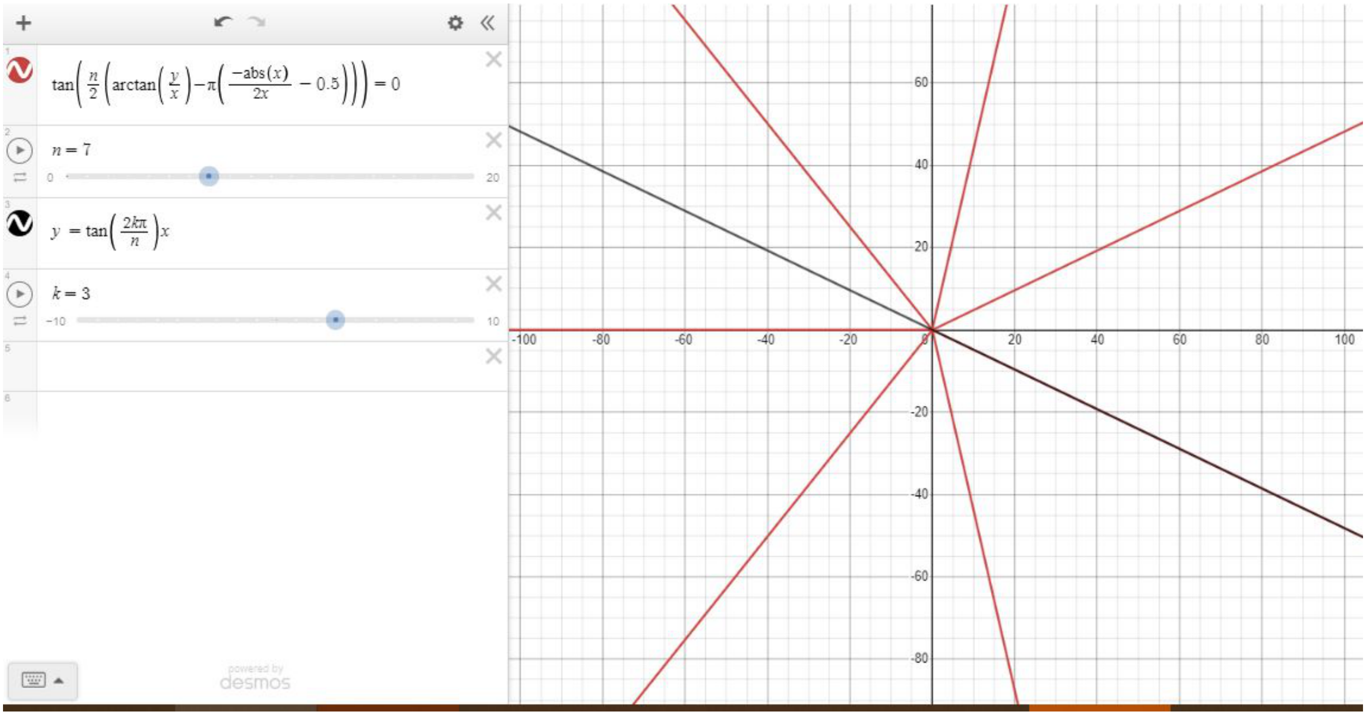
A final feature of this graph is the symmetrical spacing of petals, over the y-axis. While this is not an intended feature of my graph, it took me some time before I realized how to change it. This realization came while working on another graph.

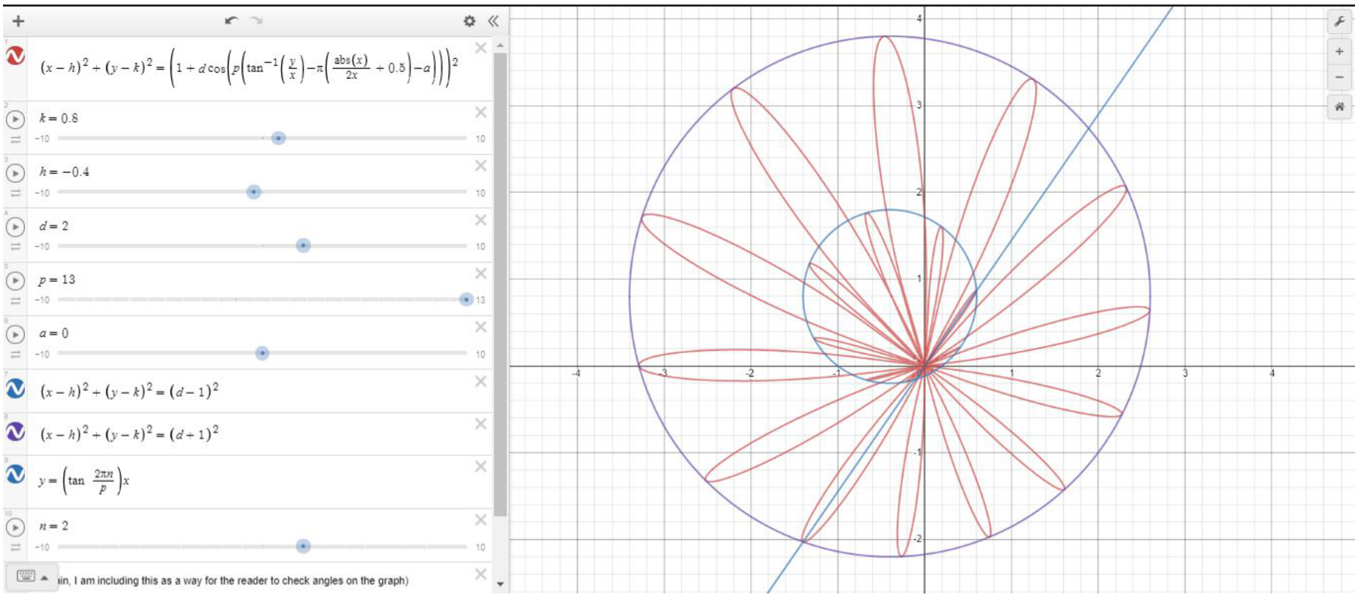
2 Rays, More Efficiently

While working with polar coordinates, I realized that it would be an interesting challenge to divide the plane into equal angles, using multiple evenly-spaced rays. Such a graph would be extremely useful for constructing other radial graphs. I approached this objective using another main advantage of polar coordinates, setting up an equation with angles as solutions. However, this can be difficult, due to the constraints of converting polar to Cartesian coordinates.

My first attempt at evenly-spaced rays extending from the origin involved the polar equation, $\tan(\frac{n}{2}\theta) = 0$. Though Desmos can't graph this, it gives a total of n solutions in the form $\theta = \frac{2\pi}{n}$, with rays on both sides of the y-axis. However, the standard way of converting polar to Cartesian coordinates involves replacing θ with $\tan^{-1}\frac{y}{x}$, which only produces the desired outputs for angles between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$. For angles outside of this range, the inverse tangent function instead gives an angle π apart from the intended angle. As a result, any attempt to convert this graph to Cartesian coordinates results in a graph where quadrants I and IV are accurate, but quadrants II and III are simply a reflection of I and IV over the origin. This is a problem for graphs with an odd number of evenly-spaced rays, but it can be fixed by graphing the rays to the right of the origin separately, and adding π to the output of $\tan^{-1}\frac{y}{x}$ so that negative angles are corrected to positive ones.

However, this addition of π didn't fully resolve the issue. While it is possible to graph every set of rays in two parts, one to the left of the y-axis and one to the right, my intention had been to graph all of my rays together. In other words, I needed a way to rotate the graph by π only in some situations, specifically when $x < 0$. Fortunately for us, the function $\frac{|x|}{x}$ produces two distinct values, -1 and 1, depending on the sign of x. By dilating this function vertically by -0.5 and translating it up, we get $y = -\frac{|x|}{2x} + 0.5$, which produces 1 if x is negative and 0 if x is positive. Rotating the value of $\tan^{-1}\frac{y}{x}$ by $\pi(-\frac{|x|}{2x}+0.5)$ results in a more accurate conversion from polar to Cartesian coordinates, so I substituted $\tan^{-1}\frac{y}{x}-\pi(-\frac{|x|}{2x}+0.5)$ for $\tan^{-1}\frac{y}{x}$ in my graph of the rays.





In conclusion, using a combination of polar and Cartesian coordinates, I continue to surprise myself with a variety of graphs. While my original goals were to create an image, preferably involving bicycles and/or flowers, for the Desmos Global Math Art Contest, I have come to see the value of creating graphs that are not just visually, but also mathematically, unique.